

Lab Exercise 05: Advanced Wireshark

BACKGROUND:

In the previous Lab, you discovered the basic properties of Wireshark. In this Lab, you will learn how to use this tool in a more advanced way to analyse network traffic.

Before starting this Lab, download **http.cap** either from

https://wiki.wireshark.org/SampleCaptures#HyperText_Transport_Protocol_.28HTTP.29 or from the blackboard. Then launch **Wireshark** and open this cap file.

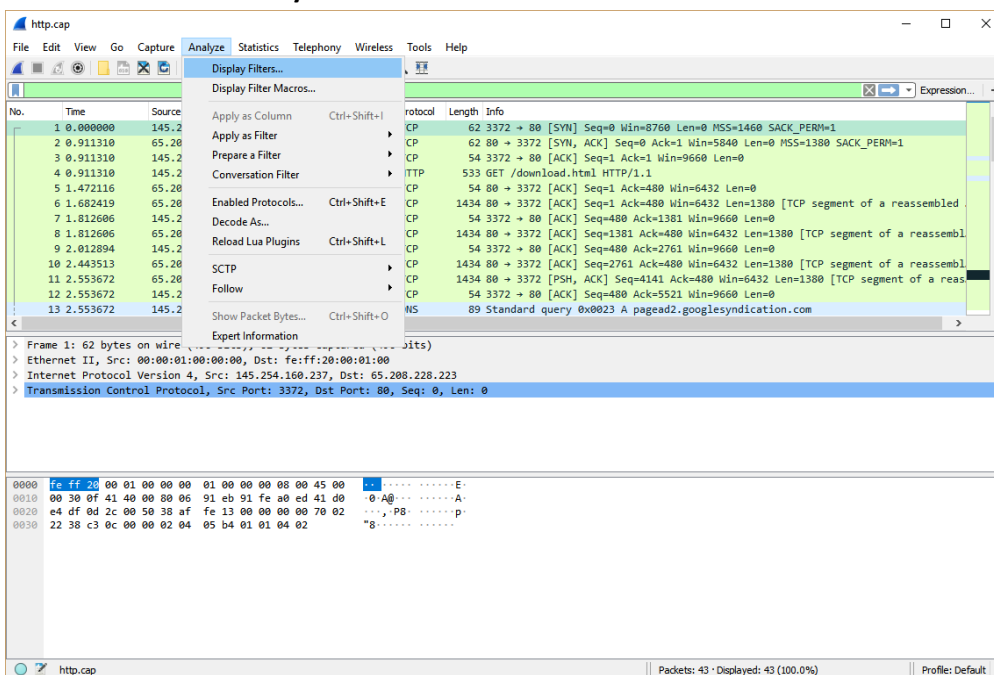
1. TRAFFIC FILTERING:

One of the most important features of Wireshark is traffic filtering. This is helpful to focus on some specific traffic. The filtering feature can be used in two ways:

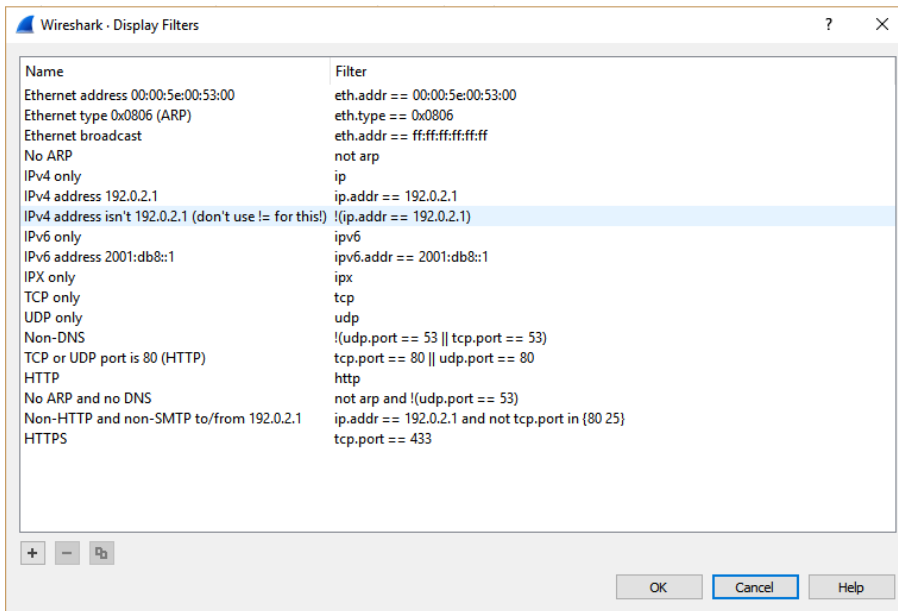
1. Offline: Filters can be applied to an already captured file. In this case, only selected packets by the criteria defined in the filter are displayed on the Packet List Pane. These types of filters are named **Display Filters**.
2. Online: Filters can be set before live capturing. In this case, only specified packets are recorded by Wireshark. These types of filters are called **Capture Filters**.

Using this feature only packets belonging to the selected traffic can be captured and displayed on the Packet List Pane. This is very useful in working on large files containing a sheer volume of packets as well as in capturing live traffics by eliminating the traffic which is not under investigation.

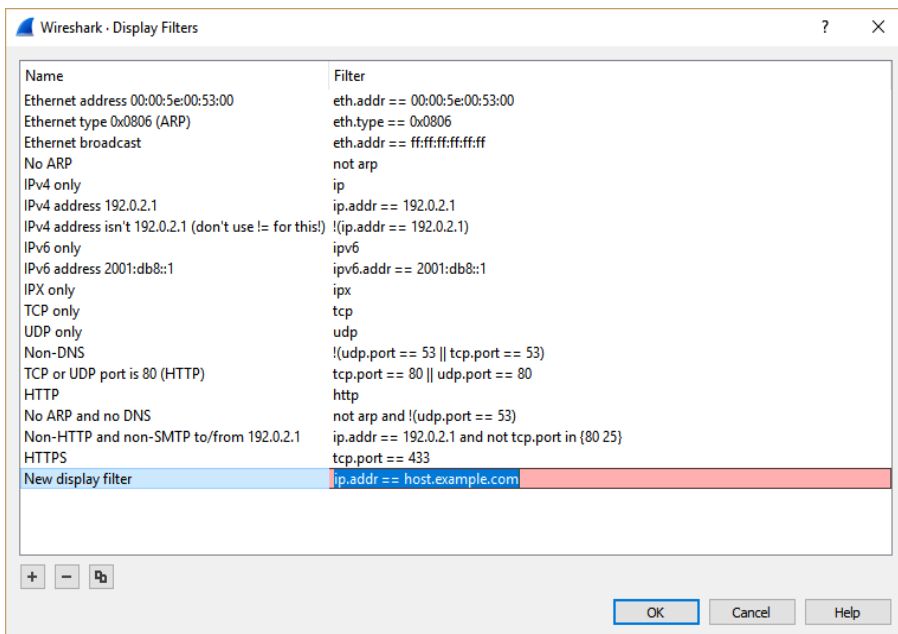
Wireshark is equipped with an extensive collection of predefined filters. You can display them using “**Display Filters**” under the “**Analyze**” menu item :



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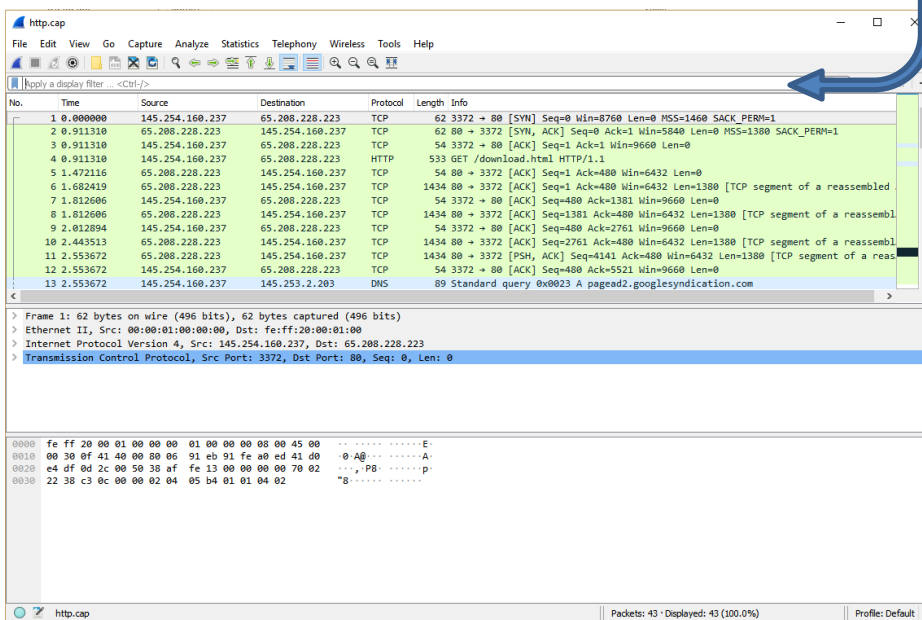


You can create a new filter and delete the existing filter using the “+” and “-” buttons:



Click on “Cancel” and proceed with the tutorial.

To use one of these filters you need either to place its name in the **"Apply a display filter"** entry field which is situated above the Packet List Pane:

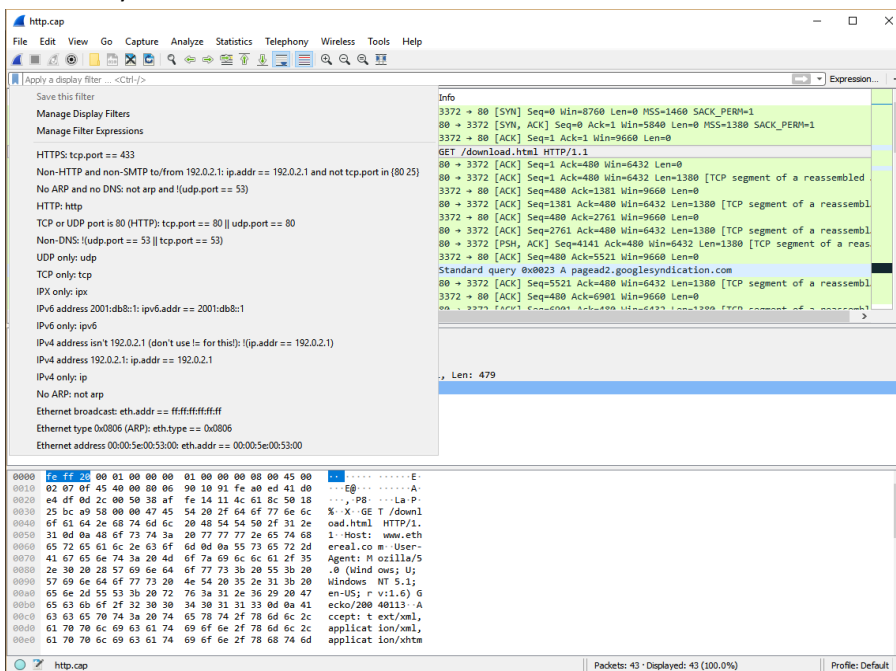


Type **http** in this field and see the effects in the Packet List Pane.

How many records displayed in the pane are there?

You may notice that when you are typing the filter Wireshark suggests filter names using its autocompleting feature. This helps you to find the appropriate handle for the filter that you want to use. You can delete the filter by clicking on the **"X"** button. This will restore the original records in the Packet Display Pane.

You can select a filter using the bookmark like button on the left hand side. This displays some of the commonly used filters:



This menu contains an option: **Manage Display Filters**. It helps you to add, remove or edit filters.

Click this option and see how you can filter **Non-DNS** traffic?

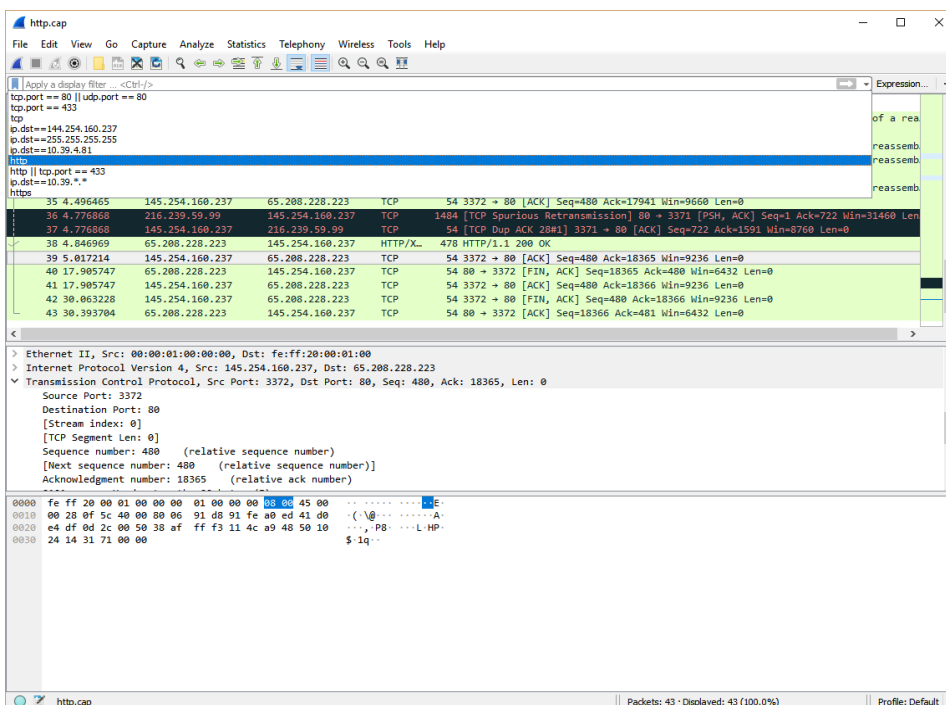
Add a filter to filter **HTTPS** traffic. Consider https is http + tls. tls uses tcp port 433. You can combine the rules using the operators && (AND), || (OR) and ! (NOT).

Type **https** in this field and see the effects in the Packet List Pane.

How many records displayed in the pane are there?

Delete **https** display filters using **Manage Display Filters**.

You can also select previously used filters by using the down arrow on the right side of the entry field. This shows a history drop-down list. Then you can easily select the filters you previously used.



You can create specific rules to filter source and destination IPs using **ip.src** and **ip.dst**.

Filter the packets whose destination is **65.208.228.223**.

Filter the packets whose source is **216.239.59.99**.

How many records have returned after each filtering operation?

Again, you can combine the rules using the operators && (AND), || (OR) and ! (NOT).

Filter packets whose destination and source are **65.208.228.223** and **216.239.59.99**, respectively.

How many records do you have?

Change the destination address to **145.254.160.237**. How many packets now?

You can add the port numbers to the filter. You can use **tcp.srcport** , **tcp.dstport** and **tcp.port** expressions for source port, destination port, and any port, respectively.

Filter all packets whose destination address is **145.254.160.237** and source or destination port is **NOT** from **80**.

2. Filtering Live Traffic

If you set a filter before starting a new traffic capture only intended packets are recorded and displayed. You can change the filter during the capture.

Filter only inbound traffic for your PC. Use your PC's IP address.

Filter only outbound traffic for your PC. Use your PC's IP address.

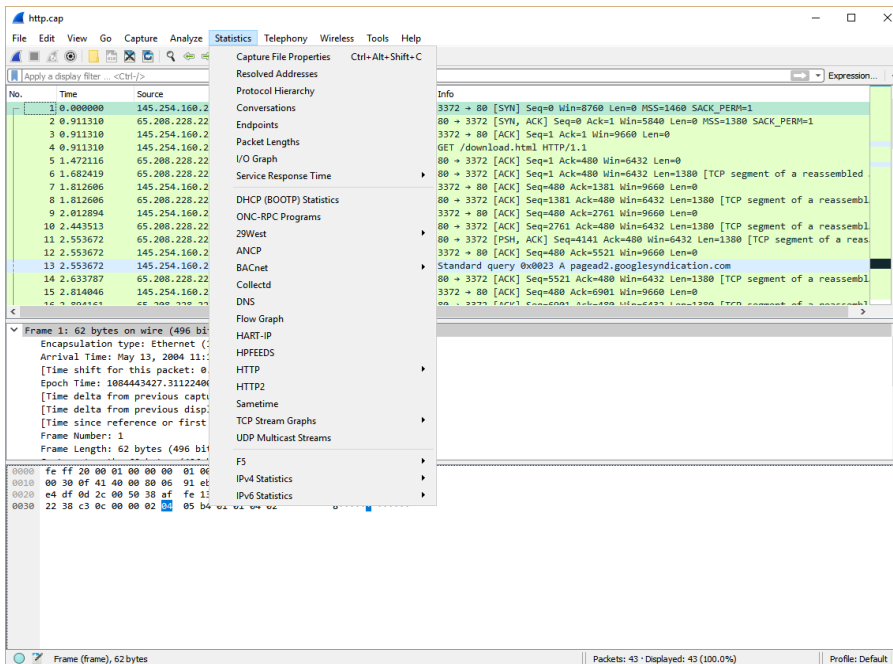
Filter the traffic which is not related to your PC. Think about the filtering rule!!

Filter the traffic which is only related to your PC. Think about the filtering rule!!

TIP: to find out PC's IP address. Open a command window and type `ipconfig /all`.

3. Statistics

One of the useful properties of Wireshark is **Statistics** Menu. This menu contains useful metrics in the **Statistics** drop-down menu to help you to analyse the traffic :



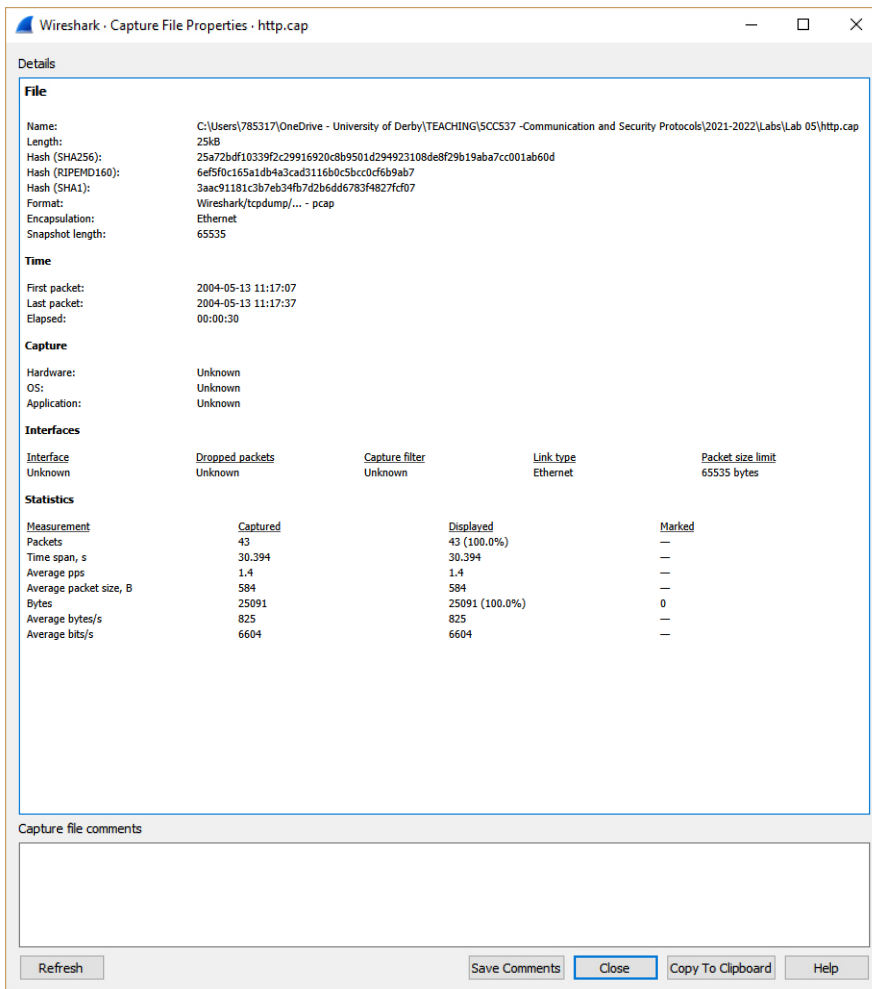
These include size and timing information about the capture file itself, along with dozens of charts and graphs ranging in topic from packet conversation breakdowns to load distribution of HTTP requests. These statistics can also be refined by display filters to highlight important information.

The network information can be simplified using the options in this menu. The important ones are :

3.1 Capture File Properties

This option displays the properties of the current capture file loaded to Wireshark. It summarizes the general statistic.

Select this option and display the properties of http.cap.



What kind of information can you gather from this screen?

You can add your comment to this capture file by typing them in the “**Capture file comments**” pane and then clicking on the “**Save Comments**” button.

Add and save your own comments for this capture file. Close the screen and reselect the **Capture File Properties** to see if your comments have been saved.

3.2 Protocol Hierarchy

This option creates a window displaying a complete table of all captured protocols. Each row contains statistical values of one protocol. They are **Percent Packets**, **Packets**, **Percent Bytes**, **Bytes**, **Bits/s**, **End Packets**, **End Bytes**, **End Bits/s**. Percent Packets and Bytes are also displayed as bar graphs. Active display filters are also displayed at the bottom.

Display Protocol Hierarchy of http.cap.

What protocols were captured in this file? Explain the protocol hierarchy.

What percentage of packets in this capture file is UDP (User Datagram Protocol)?

What percentage of packets in this capture file is HTTP?

What higher level protocols are present in the hierarchy using HTTP?

Please note that you can copy this information in either CSV or YAML format to use external applications using the **copy** button at the bottom of the window.

3.3 Endpoints

This displays a list of endpoints (a network endpoint is where protocol traffic of a specific protocol layer ends).

A network Endpoint is the logical endpoint considering separate protocol traffic of a specific protocol layer.

There are different endpoint statistics are considered in Wireshark. Some of them are as follows:

- Ethernet: Ethernet device's MAC-48 identifier.
- IEEE 802.11 : A MAC-48 address similar to Ethernet.
- IPv4 : 32-bit IPv4 address.
- IPv6 : 128-bit IPv6 address.
- TCP : A combination of the IP address and the TCP port used. Different TCP ports on the same IP address are different TCP endpoints.
- UDP : A combination of the IP address and the UDP port used, so different UDP ports on the same IP address are different UDP endpoints.

The information for different endpoints is presented in different tabs. The number of endpoints for a specific endpoint is also shown in the related tab header.

Each row in the list corresponds to a specific endpoint and show the statistical information for this endpoint. This information varies for the underlying endpoint. For example, Ethernet includes the number of packets, Bytes, number of Tx and Rx Packets as well as Tx and Rx Bytes. TCP contains port number additional to the fields included in Ethernet while IPv4 has additional Country, City, AS Number, AS Organization fields.

The displayed endpoints types can be added or removed using the “**Endpoints Types**” dropdown menu.

If it is possible, IP addresses can be replaced with their domain name using the “**Name resolution**” checkbox. The “**Limit the display filter**” checkbox is used to display the records matching with the current display filter. The “**Copy**” button is used to copy the information in CVS or YAML format.

Select the Endpoints option. Navigate between different endpoints using tabs and analyse the corresponding statistics.

How many TCP endpoints are there? What are they?

How many IPv4 endpoints? How many packets are exchanged in the IPv4 endpoint?

3.4 Conversations

This option shows the network conversation between two endpoints (For example exchange of traffic from one IP address to another). It is similar to the **Endpoints** option. This option displays additional fields like the time between the start of the capture and the start of the conversation, duration of the conversation and number of packets and Bytes in each direction.

The information for different conversations is presented in different tabs. The displayed conversation types can be added or removed using the “**Conversation Types**” dropdown menu. If it is possible, IP addresses can be replaced with their domain name using the “**Name resolution**” checkbox.

Select the Conversation option and analyse the packet exchanged between two endpoints.

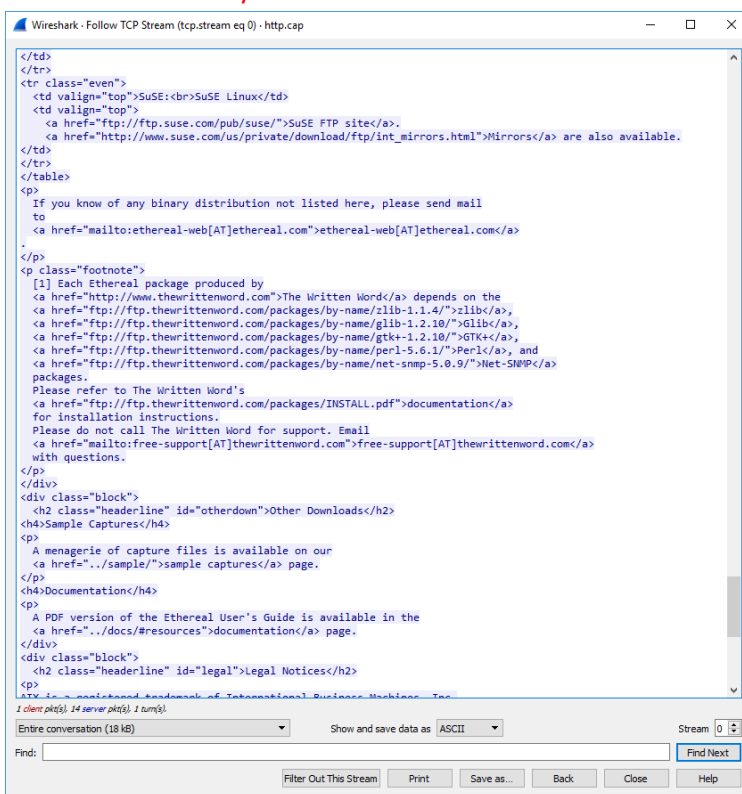
How many conversations used TCP are there?

What ports are 145.254.160.237 used for TCP communication? What are the corresponding IP addresses?

How many packets were sent to and received from 216.239.59.99?

One of the useful tools in Conversations is “**Follow Stream**” which extracts data from a TCP stream and displays it as seen by the application layer.

Select the first conversation in the TCP tab and click on “Follow Stream”. Examine the displayed data. What sort of information you can deduce?



3.5 I/O Graph

This option displays user specific graphs, visualizing the number of packets throughout the data exchange. Using this tool the number of packages can be analysed against their relative time since the capture starts or absolute time.

Select the “**I/O Graph**” option in the **Statistics** menu. The following window should be displayed:



This default representation has three graphs: One shows all packets, the other displays TCP error by applying tcp.analysis.flag filter in 1 sec interval, and another one displays filtered packets by applying tcp.stream == 0 (== can be represented by eq as well).

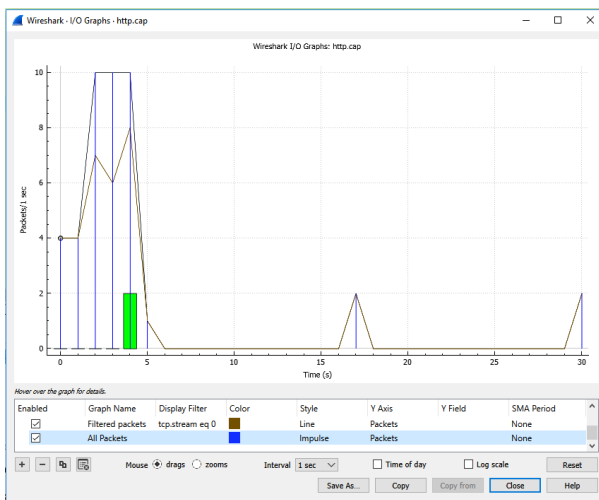
You can change the name, colour and style of these graphs as well as add/delete a filter for the graph and the information displayed in the Y axis (Packet, Byte, Bits, etc) by clicking the appropriate column. Graphs can be made visible/invisible by using the checkboxes under the Enabled column.

The time interval used in the X axis can be changed using the dropdown menu named **Interval** and the relative times used in the X axis can be replaced with actual times by checking the **Time of day**.

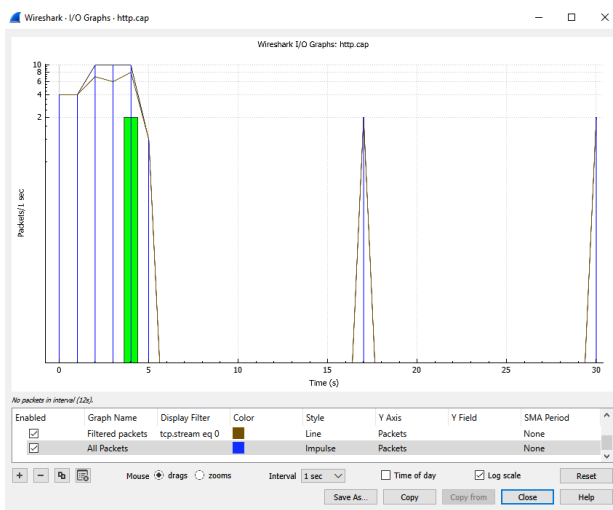
Change the colour of TCP errors to green.

The “+” button creates a new graph with default settings.

Create another graph to show the bytes of all packets using Impulse style. Set the colour to blue. Do not forget to make it visible :



As you can see the elements of packet information (1st graph) cannot be distinguished since their values are too low comparing to Byte information. In this case, using log scale by checking its checkbox makes it visible :



Check the values on the Y axis in these two windows and see the difference.

The current window displays the first 30 sec. You can see the rest of the graph by sliding the mouse in the graph by holding the left button down. The graph area can be zoomed by changing the mouse from drags to zooms. If you click the right button of the mouse you can see different operations can be carried out in the graph pane.

The current window can be saved as a pdf file using **Save as..** button.

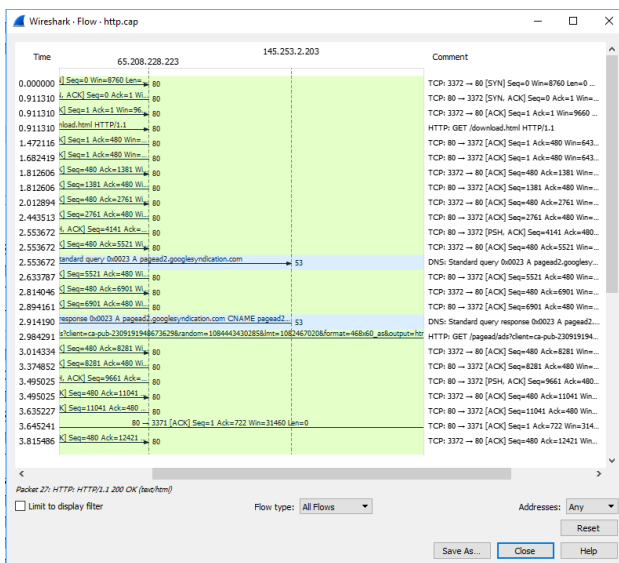
3.6 IP Addresses

To find out the underlying IP addresses can be displayed **IPv4 Statistics** and **IPv6 Statistics** under the **Statistics** menu. IP addresses can be shown in four different ways: All addresses, Destinations and Ports, IP Protocol types, Source and Destination addresses.

Display IPv4 addresses in all four ways.

3.7 Flow Graph

This shows the flow of packets in the current capture. This helps you to check connections between client and server, to find timeouts, re-transmitted frames, or dropped connections.



Display Flow Graph and examine it by using the vertical and horizontal slide bars.

Change the flow type to **TCP Flows** using the Flow Type drop down menu and examine this flow.